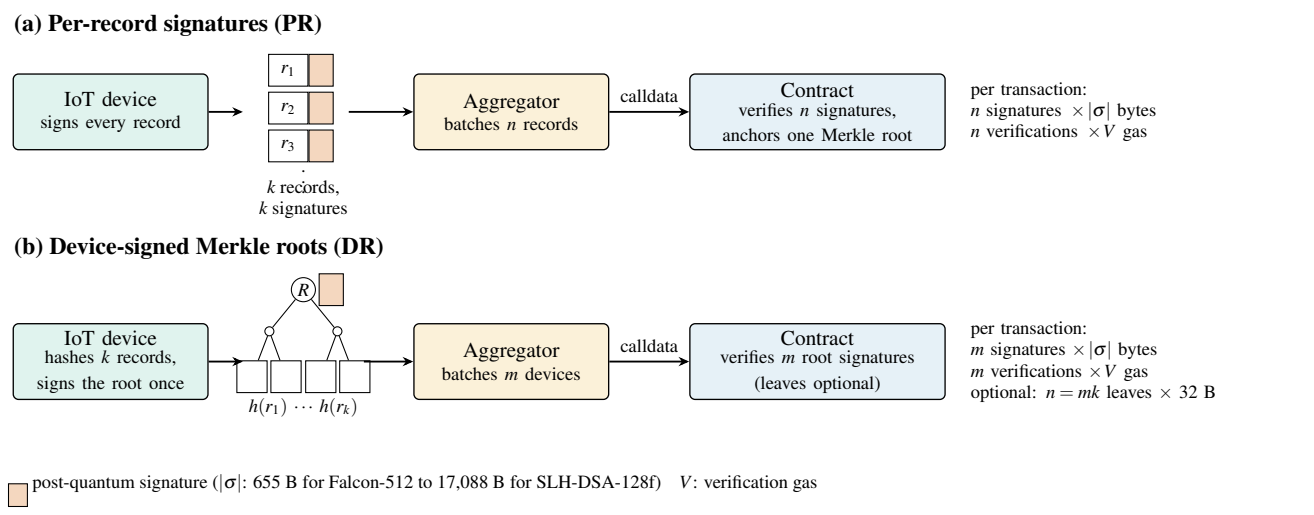

(a) Per-record signatures (PR)
IoT device
signs every record
r_1
r_2
r_3
k records,
k signatures
Aggregator
batches n records
calldata
Contract
verifies n signatures,
anchors one Merkle root
per transaction:
n signatures $\times |\sigma|$ bytes
n verifications $\times V$ gas
(b) Device-signed Merkle roots (DR)
IoT device
hashes k records,
signs the root once
R
$h(r_1) \cdots h(r_k)$
Aggregator
batches m devices
calldata
Contract
verifies m root signatures
(leaves optional)
per transaction:
m signatures $\times |\sigma|$ bytes
m verifications $\times V$ gas
optional: $n = mk$ leaves $\times$ 32 B
post-quantum signature ($|\sigma|$: 655 B for Falcon-512 to 17,088 B for SLH-DSA-128f)
V: verification gas